

Hardware Business Management System

Purpose of This Document

System components, platform feature parity, offline sync principles, and security controls. This standalone file is written for handoff to a Cloud AI, software developer, designer, database planner, or project reviewer.

How to Use This Document

- Use Confirmed requirements as agreed scope for planning and implementation.
- Use Proposed requirements as recommended starting points that can still be adjusted.
- Use Pending Decision items as questions for the business owner before coding or final schema design.
- Do not treat unresolved accounting, profit, allocation, stock costing, tax, or debt rules as final until the owner confirms them.

Status Label Convention

Confirmed means the item is part of the agreed scope. Proposed means it is a recommended implementation detail that can still change. Pending Decision means the business owner must decide before a developer or Cloud AI treats it as final.

Status	Meaning	Developer Handling
Confirmed	Part of the agreed project scope.	Implement unless later changed by the owner.
Proposed	Recommended approach or useful feature.	Validate during design and planning.
Pending Decision	Business rule or accounting detail not fully decided.	Ask the owner or make configurable. Do not hard code.

Document 04 Initial Architecture and Technical Handoff

This section gives an initial implementation direction. Architecture details are proposed unless they restate confirmed business scope. Developers may adjust the technology stack, but they should preserve the offline-first mobile requirement, central MySQL synchronization, role security, and auditability.

System Components

Component	Purpose	Status
Mobile application	Daily shop operations, offline SQLite storage, sales, purchases, expenses, debts, stock, daily opening and closing.	Confirmed
Web admin	Central administration, user and shop management, reports, settings, audit review, and oversight.	Confirmed
Backend API	Authentication, authorization, business rules, data validation, reporting, synchronization, and audit logging.	Proposed
SQLite local database	Offline storage on mobile devices.	Confirmed
MySQL central database	Central persistent data store for all shops and reports.	Confirmed
Sync service	Transfers pending mobile data to central database and retrieves approved updates from server.	Confirmed
Notification service	Delivers alerts for stock, debt, sync, daily closing, and administrative events.	Confirmed

Platform Feature Parity

Feature Area	Mobile App	Web Admin	Status
Login and role access	Yes	Yes	Confirmed
Product lookup	Yes	Yes	Confirmed
Product setup	Yes if allowed	Yes	Confirmed
Sales entry	Yes	Yes if allowed	Confirmed
Purchase entry	Yes	Yes	Confirmed
Expenses	Yes	Yes	Confirmed
Capital	Yes if allowed	Yes	Confirmed
Debts and repayments	Yes	Yes	Confirmed
Daily opening and closing	Yes	Yes for review and correction	Confirmed
Reports	Basic operational reports	Full reports and exports	Confirmed
User and permission setup	No	Yes	Confirmed
Audit review	Limited own-shop visibility if allowed	Full visibility	Confirmed
Offline mode	Yes	No	Confirmed

Synchronization Principles

- Mobile records created offline should be saved first in SQLite with a local unique identifier and pending sync status.
- When online, the app should push pending records to the backend API and receive central identifiers after successful sync.
- The backend should reject invalid, duplicate, unauthorized, or conflicting records with clear error codes.
- Each synced record should keep source device, user, shop, local creation time, server receive time, and sync status.
- Conflict resolution for the same record changed in more than one place must be decided before implementation is finalized.

Security Controls

Control	Requirement	Status
Authentication	All users must authenticate before using the system.	Confirmed

Control	Requirement	Status
Authorization	Every API and screen must check role, shop assignment, and permission.	Confirmed
Password storage	Passwords must be hashed with a modern secure algorithm.	Confirmed
Transport security	Production traffic should use HTTPS.	Confirmed
Audit logs	Critical business events must be logged.	Confirmed
Device control	Mobile devices may need registration or revocation for high-risk environments.	Proposed
Data encryption	Local SQLite encryption should be evaluated for devices storing sensitive business data.	Proposed